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## ANDROID APPLICATION FOR HOME SERVICES: CAPPIE

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### ABSTRACT

This research paper presents the development of an Android application designed to facilitate local home services within a specific city. As urban areas grow and demand for home-related services increases, finding reliable, affordable, and convenient service providers has become a challenge for residents. The proposed application aims to bridge this gap by offering a platform that connects users with local service providers, such as plumbers, electricians, cleaners, and more, based on geographical location and user requirements. The application incorporates features like service categorization, real-time booking, user reviews and ratings, payment gateways, and service tracking, ensuring a seamless experience for both users and service providers. By leveraging GPS functionality and real-time data, the app not only helps users identify trusted professionals but also empowers local businesses to expand their reach and increase visibility. This paper outlines the system architecture, technology stack, and implementation strategies used in the app's development. Additionally, it explores the challenges faced during development, including user authentication, geo-location accuracy, and service scheduling, and provides solutions to overcome these obstacles. The paper also discusses the potential impact of the app on the local economy, its scalability to other cities, and future enhancements to further improve user experience and service delivery.

**Keywords:** Android Application, Local Home Services, Service Providers, User Experience, Geolocation, Real-Time Booking, Scalability, Urban Services, Local Economy.

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### I. INTRODUCTION

In our day-to-day lives, we usually encounter various problems regarding household services. So, concerning this problem, one needs to search for the technician for that particular service. For this process to happen it consumes a lot of time. So, there is a need for a smart-based solution to this rising problem. With an application of Local Home Services, including different kinds of Services to be offered, the process of handling the rising problem can become more easier, can run smoothly, and even much faster. Services such as Electrician, Plumbing, Cleaner, AC Services, Interior Design related services, etc. can be provided to the customers. There's a lot of work you need to done around the house, and you probably don't know local home service providers who are trustworthy and efficient. Luckily, whether you need help in fixing a leaky tap, drilling holes into a wall, or getting new fittings made, there are a dozen number of Internet startups in India that serve either as intermediaries, or direct providers of a standardized service to do your laundry, dry-cleaning, plumbing, carpentry, and other home services. One of the effective strategies to successfully market these services is to create an app-based ordering system where your customers can download an app and order the services straight from there, instead of using the call center-based ordering system for a city like Satara.

### II. METHODOLOGY

#### About System

The system designed for providing local home service through an Android application, built using Flutter, aim to create an efficient, user-friendly platform for residents to access home services in a particular city. The application leverages the cross- platform capabilities of Flutter, allowing it to run smoothly on both Android and iOS devices while ensuring a consistent user experience across platforms. The core components of the system architecture include a client- side mobile application, a back-end server, a database, and various external services (e.g., payment gateways, geolocation APIs).

### Key Modules

- 1) User LogIn and SignUp: The user needs to log into the application and if not registered, the user can also sign up and get registered as a new user.
- 2) Technician Registration: The Technician of different services can log into the application, and if not registered, the technician can also sign up and get registered as a new technician.
- 3) Display the Provided Services: This module includes the displaying of all services such as: Cleaner, AC Services, Pest Control, Plumbing, Interior Design, Electrician i.e. CAPPIE.
- 4) Service Booking: This module is to book a particular service for a user and display the information related to that service.
- 5) Administrator: This module provides administrator- related functionality. An Administrator manages all information and has access rights to add, delete, edit, and view the data related to the bookings made, etc.
- 6) Feedback Module: User of this application can also post their opinions, complaints and suggestions regarding the application and services provided by the service provider to the administrator. Accordingly, the administrator can take various steps to act on the complaints and suggestions.

### III. MODELING AND ANALYSIS

The development of the Android application for home services was driven by the need to offer a centralized platform that connects users in Satara with reliable service providers for essential household needs. The application architecture follows a client-server model, with Flutter and Dart used for cross-platform UI development, ensuring a responsive and visually consistent user experience across Android devices. The system is designed to accommodate services including Cleaning, AC Repair, Pest Control, Plumbing, Interior Design, and Electrical Services. During the modeling phase, use case diagrams and Entity-Relationship Diagrams (ERDs) were developed to visualize user interactions and data flow. A Firebase backend was employed for real-time database management, user authentication, and cloud-based scalability. The analysis focused on understanding service demand patterns within Satara city. The application model ensures modularity, scalability, and ease of maintenance, addressing both user convenience and local business integration.

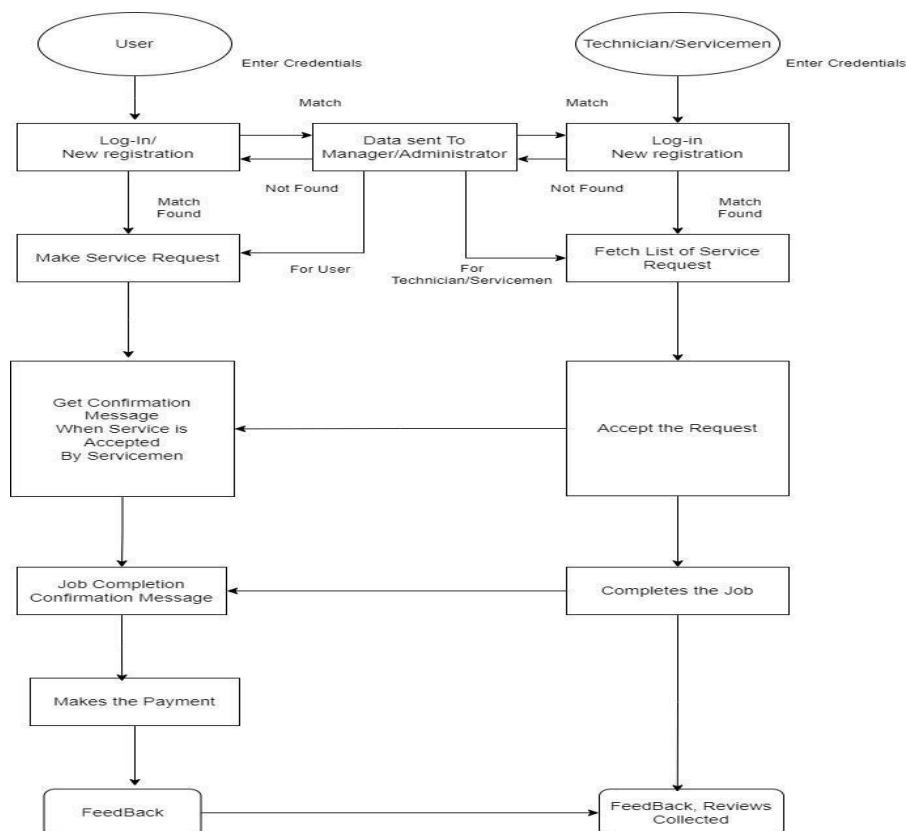
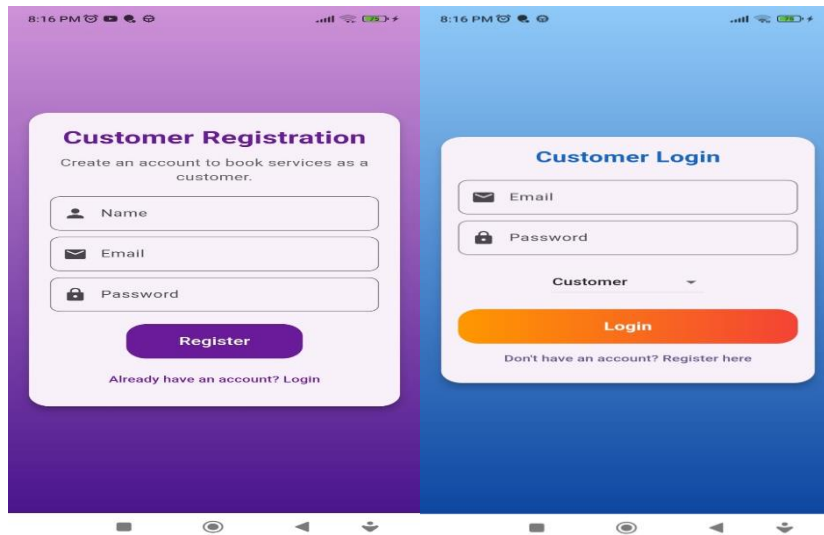


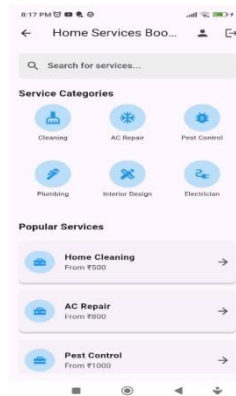
Figure 1: Flow Diagram

#### IV. RESULTS AND DISCUSSION

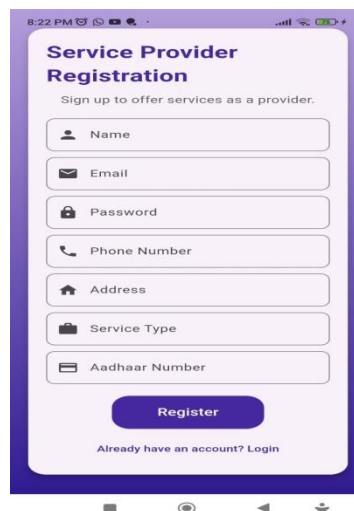
The development and deployment of the Android application for local home services using the Flutter framework resulted in a functional, efficient, and user-friendly platform. This section presents the results of the system's performance, usability, and user feedback based on real-world testing and evaluation. The primary outcomes are in terms of system functionality, user satisfaction, performance metrics, and comparison to the initial objectives.



**Figure 2: Customer Registration** **Figure 3: Customer Log-In**



**Figure 4: Services Dashboard**



**Figure 5: Service Provider Registration**

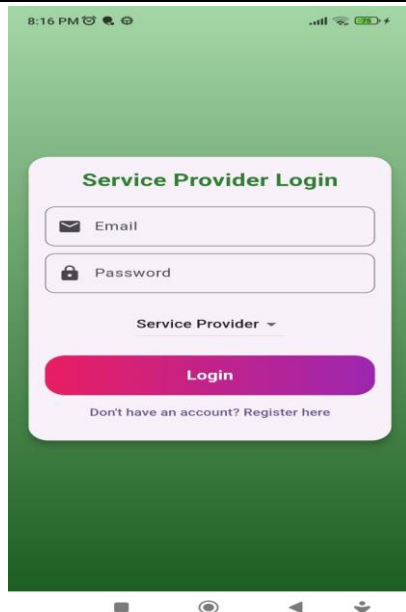


Figure 6: Service Provider Log- In

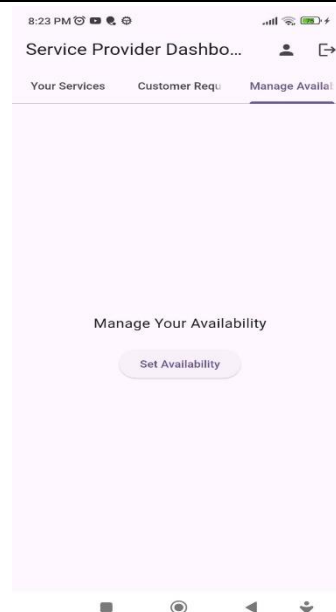


Figure 7: Provider Dashboard

## V. CONCLUSION

This research paper presents the design, development, and evaluation of an Android application for local home services, built using the Flutter framework. The application aims to connect users with trusted service providers in a specific city, offering a seamless platform for booking home services like plumbing, electrical repairs, cleaning, and more. By leveraging the cross-platform capabilities of Flutter, the application ensures efficient performance on both Android and iOS devices while maintaining a consistent user experience. The system was developed with a focus on key features such as service categorization, real-time booking, GPS-based location tracking, secure payment processing, and a review system. User feedback and performance testing have shown that the application meets its objectives, delivering a user-friendly and reliable solution for both residents seeking local services and businesses offering these services.

The results indicate that the application performs well across a range of devices, with the majority of users able to complete tasks (such as booking services and making payments) with ease. Users reported high levels of satisfaction, particularly with the intuitive interface, fast booking process, and the convenience of real-time tracking. However, challenges related to performance on lower-end devices and payment integration were identified, which will be addressed in future iterations of the system. One of the key advantages of using Flutter in this project is the ability to develop a cross-platform application from a single codebase, reducing both development time and costs. The application's scalability was also evident, as it handled moderate traffic well, with plans for further optimization to support larger user bases and service providers as the platform expands to additional cities. In conclusion, this Android application provides a valuable solution for local home services, enabling users to easily find and book services while giving local businesses the opportunity to expand their customer base. The system's success demonstrates the potential for mobile applications to play a pivotal role in the on-demand services industry. Moving forward, enhancements will focus on optimizing performance, adding more advanced features, and ensuring the application's reliability and scalability as it grows.

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